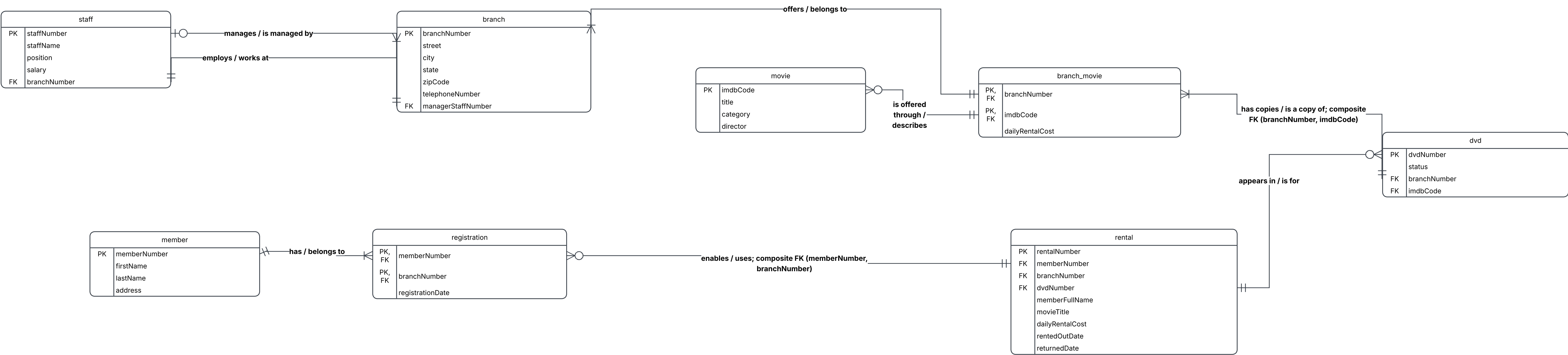


DVD Rental Company ER Model



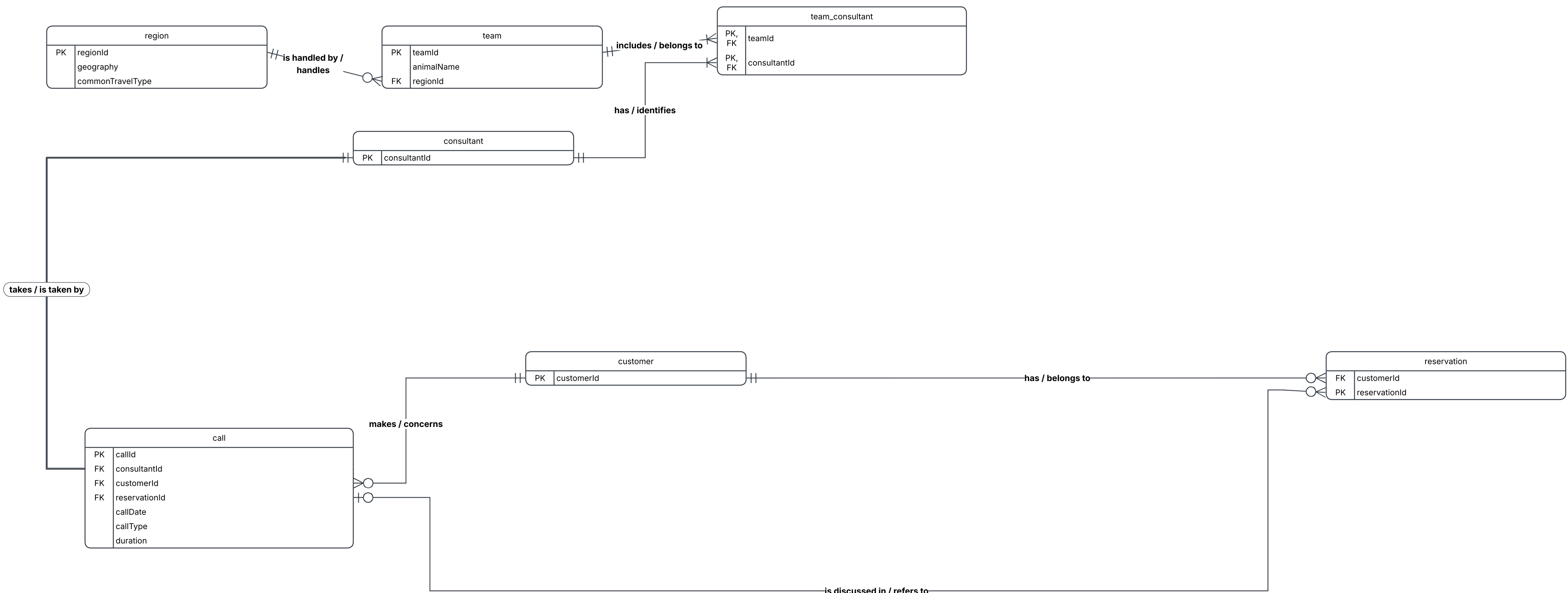
Assumptions / Constraints

- A member may rent from a branch only if the member has a registration for that branch.
- The rented DVD must belong to the same branch as the registration used for the rental.
- A member may have at most 10 active rentals at one time across the company. An active rental has no returnedDate.
- One DVD copy may have at most one active rental at a time, though it can have many past rentals.
- A branch has exactly one current manager, and that manager must be staff currently assigned to the branch; managerStaffNumber must reference staff assigned to that same branch.
- returnedDate is blank until the DVD is returned.
- Staff reassignment and manager changes describe the current assignment; historical staffing assignments are outside this model.
- Example movie categories: Action, Adult, Children, Drama, Horror, and Sci-Fi. No separate category entity is modeled.

Composite keys: branch\_movie(branchNumber, imdbCode) and registration(memberNumber, branchNumber). Composite foreign keys are shown on rental and dvd.

Entity/table names are singular lower-case snake\_case; attribute names use camelCase. PK = primary key; FK = foreign key.

# Travel Consulting Company ER Model



## Assumptions / Constraints

- The diagram shows each team's current region. Team reassignment is represented by changing its regionId; historical region assignments are outside the stated requirements.
- Consultants may be assigned to multiple teams at the same time.
- reservationId is required when callType is "existing reservation" and blank when callType is "new travel need."
- If a call refers to a reservation, that reservation must belong to the customer identified on the call.
- Daily call counts by type, durations, and monthly statistics are calculated from logged calls. No separate statistics table is modeled.
- A new travel need is represented by the call's type; no separate travel-need entity is needed because no details about a travel need are specified.
- The number 12 is a company rule for the number of teams, not a cardinality between team and region.

Table names are singular lower-case snake\_case; attributes use camelCase. PK = primary key; FK = foreign key.